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Intro to Data Science 01:198:439
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Predicting EV Charging Infrastructure Needs in New Jersey

[Github](#)

[Demo Video](#)

1. Project Statement

The goal of this project is to understand whether New Jersey has enough public charging infrastructure to support the increasing number of electric vehicles. Adoption of electric vehicles continues to rise because of clean energy incentives, lower operating costs, and a growing interest in sustainable transportation. However public charging access is not expanding evenly across regions. Some counties have more stations and easier access while others have very limited coverage. This creates challenges for drivers who rely entirely on public charging including renters, students, and people who do not have home charging options.

My objective was to analyze public charging availability together with demographic characteristics and then use these attributes to build a simple predictive model. I began by collecting charging station locations, electric vehicle registration records, and demographic features from state and federal sources. At first I planned to build a model focused on New Jersey counties. After reviewing the data it became clear that this approach was not possible. Many counties had only zero or one confirmed public station and electric vehicle registration data was only available at the state level. With such limited variation a county model would not be statistically meaningful.

Because of this I expanded my analysis to all fifty states. This allowed me to use complete and reliable datasets while still answering my main question about how New Jersey compares. I chose not to create artificial county values for New Jersey because synthesizing data would weaken the credibility of the results. Since accurate federal level data already existed the better choice was to work only with real observations and maintain the integrity of the project.

2. Novelty and Importance

Electric vehicle infrastructure is expanding but not at the same pace in every region. Many reports summarize the number of electric vehicles or the number of charging stations but very few combine demographic patterns, geographic structure, and adoption behavior in one analysis. This project is important because it brings these elements together and evaluates how they relate to one another. By studying population, income, renter share, commute time, and

population density alongside charging station counts, I was able to identify patterns that would not appear if each variable were viewed alone. This work is also personally meaningful because I have been researching electric vehicles for my own use and this project helped me understand how infrastructure decisions affect real communities and how they influence future adoption.

3. Progress and Contribution

I began by cleaning and preparing all datasets. I standardized location names, removed invalid values, and created new features including chargers per one hundred thousand residents. I performed exploratory analysis to understand how these features interact and which attributes might influence charging availability. When I studied New Jersey counties separately, I confirmed that there was not enough usable information to build any valid model. The county charging counts were almost entirely zero or one and electric vehicle registration counts at the county level were unavailable. This reinforced the need to expand the project beyond a single state.

After shifting to a nationwide analysis I created a complete modeling dataset that brought together charging counts and demographic attributes for all states. I organized my code into a simple and clear pipeline which followed the general workflow from our class labs. This included collecting the raw data, cleaning and merging it, analyzing it visually, and finally building predictive models.

4. Models and Algorithms

I tested linear regression and random forest regression. Linear regression was chosen because it is interpretable and aligned with the methods we learned in lecture. It helped me understand how each feature influences charging availability and whether the relationships were positive or negative in a straightforward way. Random forest served as a second method that allowed for nonlinear relationships and interactions that a simple linear model may not capture. Both models used the same predictors which included population, median household income, renter occupied share, mean commute time, and population density. The target variable was chargers per one hundred thousand residents. This measure provides a fair comparison across states that vary widely in size.

5. Experimental Design

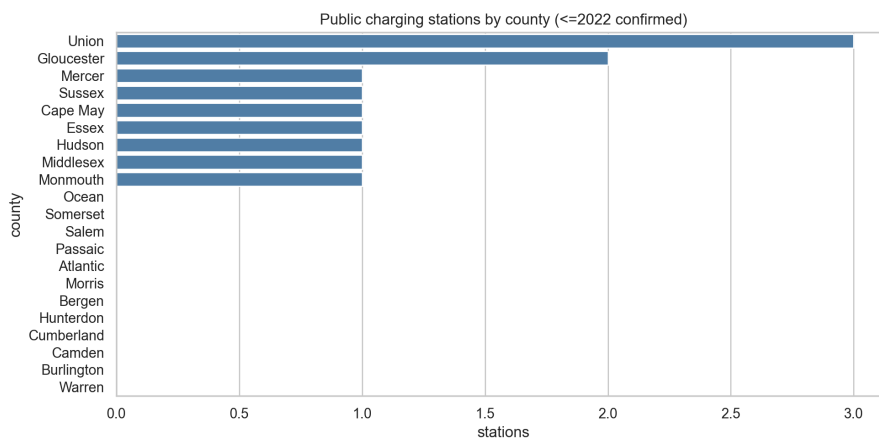
I created the final modeling dataset and standardized the predictors. I used an eighty percent training split and a twenty percent testing split. The evaluation metrics were root mean squared error and R squared. R squared is helpful in this context because it shows how much of the variation in charging availability is explained by the demographic features which makes it easier to understand the overall effectiveness of the model. These metrics are consistent with the

techniques we used in our class labs. Although the dataset contained only fifty observations there was enough variation across states to produce meaningful patterns.

6. Results and Evaluation

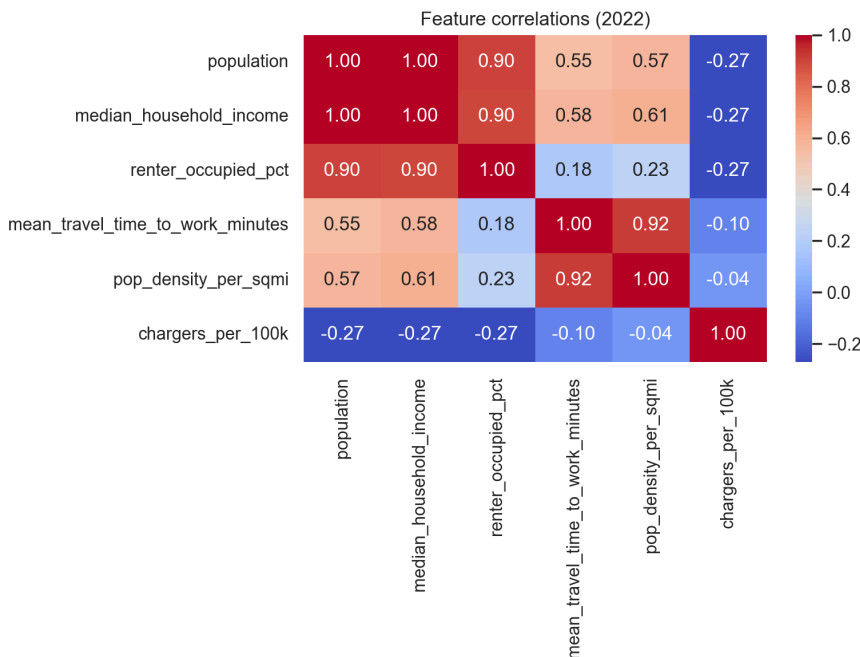
The analysis began at the county level for New Jersey. I created exploratory visualizations to understand whether the data had enough variation to support a valid model. These visualizations appear alongside the state level versions because the project required a shift in scope. Presenting both sets is important because it shows what I originally attempted and what eventually became feasible.

Figure vis_nj_bar.png



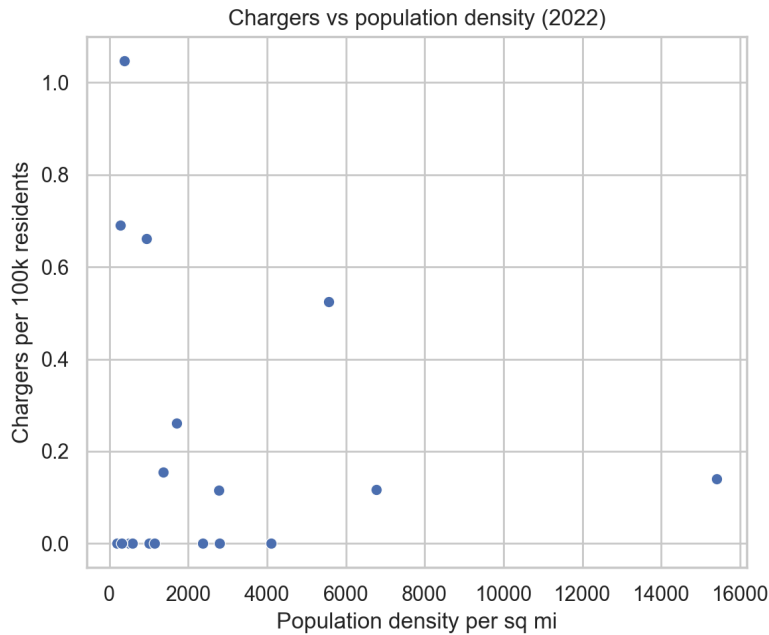
From this figure I learned that only nine counties in New Jersey reported at least one confirmed public charging station. Most counties had zero stations. This lack of variation made it clear that no predictive model would behave reliably. It also helped explain why the county model produced extremely unstable results.

Figure vis_nj_heatmap.png



From this figure I learned that New Jersey county features do not display meaningful correlations with charging access. The values are dominated by zeros and single station counts which prevents any relationship from appearing. This confirmed that the dataset does not support a statistical model at the county scale.

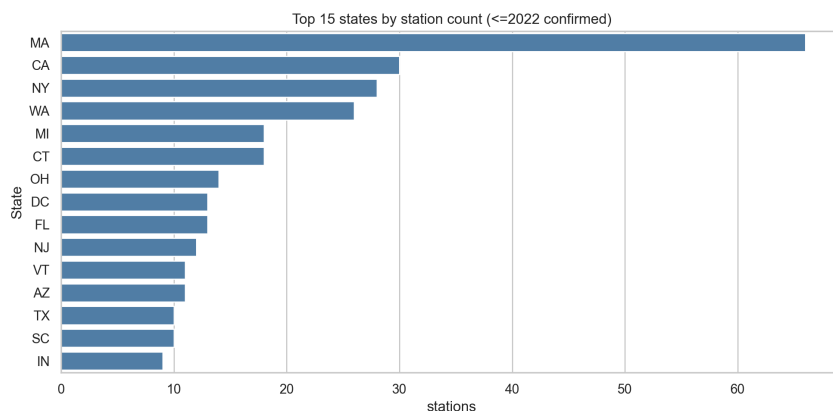
Figure vis_nj_scatter.png



From this figure I learned that chargers per capita do not form any pattern across counties. All points collapse into a horizontal cluster at or near zero. This shows that the response variable does not contain usable variation for prediction.

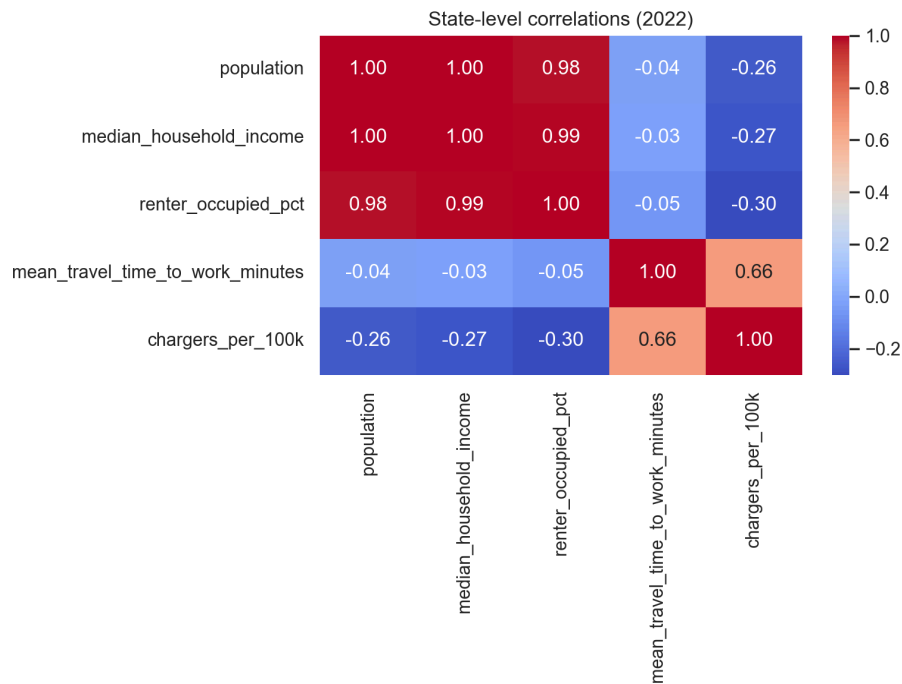
These county level findings explain why the New Jersey model produced negative R squared values such as negative four hundred seventy four point eight one six for linear regression and negative eighteen point one three two for random forest. These results show that the model performed worse than simply predicting the mean. This outcome is expected when the target variable does not vary. Because of these issues I expanded the analysis to all fifty states. The state level data contains consistent demographic values for every state and meaningful differences in charger availability.

Figure vis_state_bar.png



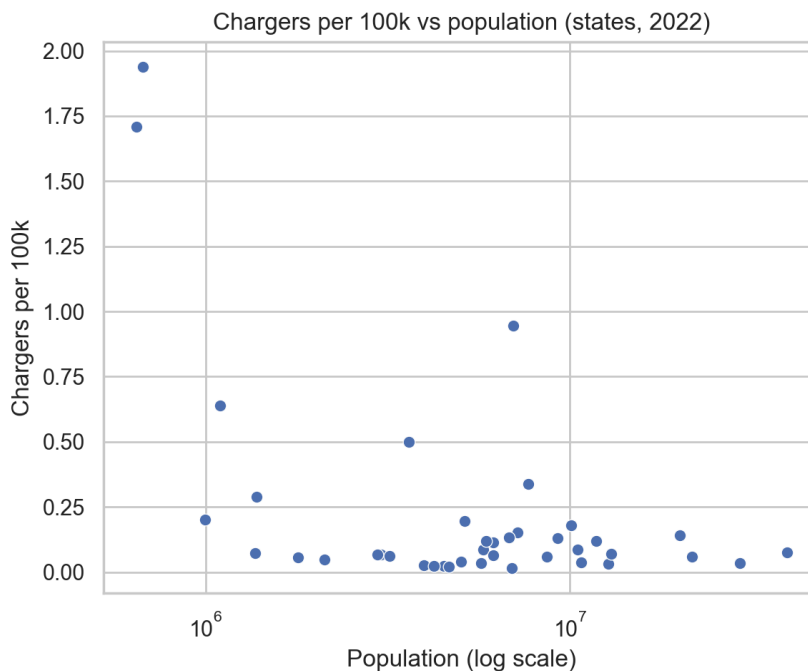
From this figure I learned that some states such as Massachusetts, Washington, and California have significantly higher station counts compared with the rest of the country. New Jersey falls in a mid level range which helps contextualize its infrastructure needs.

Figure vis_state_heatmap.png



From this figure I learned that demographic factors such as population density and commute time show recognizable patterns with chargers per capita at the state level. Unlike the New Jersey county version this heatmap shows meaningful variation.

Figure vis_state_scatter.png



From this figure I learned that states with very large populations tend to have fewer chargers per one hundred thousand residents. This reinforces the importance of standardizing charging availability by population. The state level model produced stable and interpretable results. Linear regression achieved an RMSE of zero point three one seven and an R squared of zero point six five seven. The random forest model achieved an RMSE of zero point three two six and an R squared of zero point six three eight. These values show that the predictors explain a meaningful portion of the variation in charging access.

Section 7. Advantages Limitations and Changes From Proposal

The original goal of this project was to build a New Jersey county level model. The county data proved to be the strongest limitation. Only nine counties had any confirmed public charging stations and even these had counts of one or two. Electric vehicle registrations were not available at the county level which prevented me from calculating ratios such as chargers per vehicle. Because of this the model had almost no variation to learn from and produced negative R squared values. These results were expected given the input data and they demonstrated why continuing with this approach would not produce meaningful insights.

I also considered the idea of generating synthetic county values. I chose not to do this because generated data would introduce artificial behavior that does not reflect real infrastructure conditions. Since accurate federal data already existed at the state level it was more responsible to build the model using real observations rather than created numbers. This led to the most important change from the proposal. Instead of continuing with the county model I expanded the analysis to every state. This change strengthened the project because it provided enough variation for real statistical learning while still answering the question of how New Jersey compares. It also reflects a common data science pattern where the original plan must evolve once the data is examined.

I also learned that some of the missing information in the New Jersey county dataset may be due to real events that affect how infrastructure is recorded. A recent decision by the New Jersey Turnpike Authority required the removal of multiple Tesla Supercharger sites on the Turnpike, which means that public data sources may not show chargers that once existed or were still in transition. This helped me understand that the absence of chargers in the federal dataset does not necessarily mean that no chargers were present. It simply means that the chargers were not included in the specific federal feed used in this project. I also discovered state level dashboards like EValueNJ which track detailed charging information for New Jersey. These dashboards suggest that complete county level data does exist, but it is not publicly released in a structured form that could be used within the scope of this class. This confirmed that the limitations I encountered were not due to a mistake in my process but were tied to how different organizations define and share charging infrastructure data.

8. Conclusion and Future Work

This project helped me understand how the availability of data directly shapes the kind of analysis that is possible and how important it is to adjust the approach when the initial plan does not match the reality of the dataset. The New Jersey county analysis showed that even when a topic is relevant and meaningful it cannot be modeled if the underlying data does not contain enough variation. Almost every county had zero confirmed stations and no county level registration data existed. This revealed that the strength of any model depends on the strength of its inputs and that a clean dataset is not enough if the values themselves do not change in a meaningful way. This was one of the most important lessons in the project.

The state level analysis showed the opposite pattern. Once I worked with a dataset that contained complete values for every region and meaningful differences between states the modeling process became informative. I learned how demographic characteristics influence public charging availability and how states compare with one another. This helped me understand that public charging access is shaped by a combination of density, commuting behavior, urban structure, and population size. These patterns gave context to New Jersey and explained why its charging network may require expansion to keep pace with adoption.

I also learned how predictive modeling behaves in small datasets. With only fifty observations the model must be interpreted carefully but it can still reveal meaningful relationships when the features are well chosen and consistently measured. This reinforced the importance of using metrics such as RMSE and R squared to evaluate whether the model captures real structure in the data. If I continue this work I will focus on obtaining county level registration counts and more precise information on private and workplace charging stations. Additional years of data would allow me to observe how infrastructure is developing over time. I would also explore geographic clustering methods to understand which regions are underserved in relation to nearby areas rather than only within state boundaries. Together these steps would create a more complete picture of charging access and support more detailed planning.

While working through this project I also saw how real world policy decisions can change data availability in ways that directly affect modeling outcomes. The Turnpike decision to remove Tesla Superchargers and the existence of detailed state dashboards that are not fully accessible showed me that data gaps are not always technical issues. They can come from ownership choices, reporting rules, and the way different agencies define what counts as public infrastructure. This helped me frame my results in a more informed way and reminded me that successful modeling depends as much on understanding the data environment as it does on the code itself.

Overall this project taught me how data cleaning, exploratory analysis, feature engineering, and model building connect to one another. It also showed me that data science always begins with understanding the dataset and sometimes the most valuable outcome is learning why an early idea cannot work. By adjusting the scope and focusing on the strongest available data I was able to create a model that provides clear insights into charging infrastructure needs and gives a realistic view of how New Jersey compares to other states as electric vehicle adoption continues to grow.

References

- New York Times. (2025, June 3). *Tesla chargers removed from New Jersey Turnpike sites after policy shift*.
<https://www.nytimes.com/2025/06/03/nyregion/tesla-chargers-new-jersey-turnpike.html>Atlas
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